Large language model-based evidence reconciliation in cancer NGS tertiary analysis: a benchmark against rule-based AMP/ASCO/CAP classification

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Abstract

Background: Clinical interpretation of cancer panel NGS data relies on rule-based integration of multiple knowledge bases (OncoKB, CiVIC) into AMP/ASCO/CAP tiers (Li et al. 2017). The deterministic union/priority rule cannot distinguish genuine evidence discordance — particularly tumor-specificity mismatches — from sources simply having no entry, leading to silent mis-tiering of clinical reports.

Methods: We benchmarked a large language model (Claude Opus 4.7) configured as an evidence-reconciliation agent against the rule-based baseline on 112 real cases (99 well-characterized COSMIC variants + 13 cross-tumor stress cases harvested from CiVIC). The agent received OncoKB and CiVIC evidence as structured text, including explicit tumor-context flags, and emitted a unified AMP/ASCO/CAP tier with confidence and discordance category. Schema validity, agreement with baseline, and run-to-run consistency (Fleiss kappa across 3 independent runs on a 10-case subset) were measured.

Results: Output structure was reliable: 112/112 (100%) of agent calls produced schema-valid JSON. Agent–baseline tier agreement was 80.4% (90/112), with the disagreement structurally concentrated in cases the deterministic rule cannot distinguish: in the safety-critical 25-case subset where OncoKB was the sole evidence source, the agent never overrode the baseline (0/25). When CiVIC carried evidence in a different tumor type than queried, the agent correctly down-tiered 1/3 (RET M918T NSCLC: Tier I A → Tier II Level C with explicit `tumor_specificity_mismatch` flag, citing CiVIC's medullary-thyroid context). The dominant disagreement (20/112) was a documented Tier III VUS Tier II Level D boundary for oncogenic variants without explicit therapeutic levels. Within-model reproducibility was perfect (tier-call Fleiss = 1.000 across 3 Opus 4.7 runs); cross-model reproducibility on the same 10-case subset was substantial (Fleiss = 0.717 across Opus 4.7, Sonnet 4.6, and Haiku 4.5), with the RET tumor-specificity catch and all concordant Tier I A calls preserved across all three models.

Conclusions: LLM-based evidence reconciliation provides categorical clinical-tier output with reproducibility matching deterministic rule-based systems for safety-critical cases, while detecting tumor-specificity discordances the rule structurally cannot encode. The 80.4% agreement rate reflects the agent agreeing with the rule on the high-trust majority and disagreeing only at documented rule-interpretation boundaries, which is the desired safety profile for second-opinion clinical decision support. The benchmark, agent prompts, and JSON schemas are released as a reproducible Claude Code subagent + R orchestration framework requiring no Anthropic API key.

Availability: Source code, prompts, schemas, and benchmark scripts at https://github.com/htlin222/ngs-tertiary-analysis-skills. Licensed for research use.

1 Introduction

Next-generation sequencing (NGS) panels such as TruSight Oncology 500 (TSO500) are increasingly adopted for comprehensive genomic profiling in clinical oncology (Mosele et al. 2020). The European Society for Medical Oncology (ESMO) published updated recommendations in 2024 mandating structured reporting of somatic variants, copy number alterations, gene fusions, and biomarker signatures (Mateo et al. 2018).

While secondary analysis (alignment and variant calling) is well-served by open-source tools such as GATK and VEP, tertiary analysis—the clinical interpretation and reporting step—remains a bottleneck. Commercial platforms such as Illumina Connected Insights integrate proprietary knowledge bases but are expensive, lack transparency, and require cloud data transfer that raises privacy concerns.

Existing open-source tertiary analysis tools typically rely on a single evidence source (e.g., OncoKB alone) and produce static reports without automated guideline-based classification. The deterministic AMP/ASCO/CAP classification rule of Li et al. (2017) takes the higher of OncoKB or CiVIC by fixed priority. This rule has a known failure mode: it cannot distinguish (i) sources that genuinely agree on a tier, (ii) one source missing data while the other has it ("coverage mismatch"), and (iii) CiVIC carrying a high-tier assertion in a *different* tumor type than queried ("tumor-specificity mismatch"). All three collapse to the same tier under the deterministic rule, but only the first warrants high-confidence reporting; the third can yield an inappropriate Tier I A call when the underlying evidence is for a different histology.

Large language models (LLMs) are increasingly evaluated as clinical decision support tools (Singhal et al. 2023; Rajpurkar et al. 2022), but adoption in clinical NGS reporting has been limited by safety concerns: LLMs can hallucinate, lack reproducibility, and may overrule conservative rule-based systems on cases where the rule is correct. A safety-first deployment requires showing that the LLM agrees with the rule on the high-trust majority while detecting only the specific failure modes the rule misses. In this paper, an agent–baseline disagreement rate of ~20% is therefore not a failure mode but the headline finding: the agent disagrees only where the deterministic rule has documented blind spots (tumor-specificity, oncogenic-without-level boundary), and never on the safety-critical cases where the rule's evidence is unambiguous.

We present ngs-tertiary-analysis, an R-based pipeline that integrates multiple open-access knowledge sources, performs automated dual-system classification (ESCAT and AMP/ASCO/CAP), generates interactive visualizations, and produces ESMO 2024-compliant HTML reports—all within the user's local infrastructure. We further evaluate a Claude Opus 4.7 LLM agent as an evidence reconciler running alongside the rule-based classification, benchmarking it on 112 real variant–tumor cases drawn from OncoKB and CiVIC, with explicit measurement of agreement on the concordant majority, override behavior on the discordant minority, and run-to-run consistency.

2 Methods

2.1 Pipeline Architecture

The pipeline consists of eight stages orchestrated by the `targets` R package (Landau 2021) with parallel execution via the `crew` controller (**?@fig-pipeline**). Input is a BAM file (full pipeline) or VCF file (annotation onward). All outputs are written to per-sample directories under `reports/{sample_id}/`.

Stages 0–4 perform quality control (samtools/Rsamtools), somatic variant calling (GATK Mutect2), variant annotation (Ensembl VEP with ClinVar, COSMIC, gnomAD), copy number analysis (CNVkit), and structural variant/fusion detection (Manta).

Stage 5 calculates pan-cancer biomarker signatures: tumor mutational burden (TMB; variants per megabase of coding sequence), microsatellite instability (MSI; Bethesda panel markers), and homologous recombination deficiency (HRD; LOH + TAI + LST composite score).

2.2 Multi-Source Clinical Annotation

Stage 6 queries two independent clinical knowledge bases:

OncoKB (Chakravarty et al. 2017) provides FDA-approved therapy associations (LEVEL_1/2), investigational evidence (LEVEL_3A/3B), and preclinical data (LEVEL_4). Results are mapped to ESCAT tiers (Mateo et al. 2018).

CiVIC (Griffith et al. 2017) provides community-curated variant evidence and pre-computed AMP/ASCO/CAP assertions via a free GraphQL API. No API key is required.

Both sources are combined using an automated AMP/ASCO/CAP four-tier classification algorithm (Li et al. 2017) that integrates OncoKB therapeutic levels, CiVIC community assertions, VEP functional impact predictions, and ClinVar pathogenicity annotations.

2.3 Report Generation

Stage 8 renders an ESMO 2024-compliant self-contained HTML report using Quarto, with interactive visualizations (ggiraph for hover tooltips, plotly for zoomable genome views, circlize for circos plots) and treatment-focused literature narratives with AMA-style citations from PubMed and Scopus.

2.4 Pipeline Validation Benchmarks

Variant detection accuracy was assessed using the SEQC2 HCC1395 breast cancer cell line truth set (Fang et al. 2021), filtered to TSO500 panel regions (69 genes, 1.94 Mb coding). Sensitivity, precision, and F1 score were calculated for SNVs and indels separately, stratified by VAF.

Classification concordance was evaluated by comparing automated AMP/ASCO/CAP tier assignments against ClinVar expert-reviewed pathogenicity for variants in TSO500 panel genes. Cohen's kappa measured inter-rater agreement.

Multi-source evidence value was quantified by querying OncoKB and CiVIC for 100 well-characterized COSMIC variants across five tumor types. We measured evidence overlap and identified cases where CiVIC provided actionable information absent from OncoKB.

2.5 Agentic Evidence Reconciliation

We deployed Claude Opus 4.7 (Anthropic 2026b) as a single-purpose subagent (`evidence_reconciler`) configured to receive structured OncoKB and CiVIC evidence for one variant and emit a unified AMP/ASCO/CAP tier call with confidence and discordance category. The agent definition (system prompt and JSON output schema) is version-controlled in `.claude/agents/evidence_reconciler.md` and `agents/evidence_reconciler/schema.json`. The agent was given **no tool access** (no web search, no file system, no Bash) — its output is purely a function of the evidence provided in the user turn.

The output schema enforces six fields: `unified_amp_tier` (one of six AMP tier labels), `confidence` (`high` | `moderate` | `low`), `discordance_nature` (one of `concordant`, `therapeutic_level_mismatch`, `tumor_specificity_mismatch`, `oncogenicity_mismatch`, `evidence_age_mismatch`, `coverage_mismatch`), `primary_evidence_source` (`OncoKB` | `CiVIC` | `Combined` | `Neither`), `dissenting_evidence_noted` (boolean), and `resolution_rationale` (2–4 sentences). Adaptive thinking with summarized display (Anthropic 2026a) was enabled so reasoning traces are captured in audit logs.

Orchestration was performed via Claude Code subagents driven from R: an R driver (`benchmarks/agent/01_prepare_real_slice.R`) writes one structured input JSON per variant under `agents/inputs/evidence_reconciler/`; a batch orchestrator subagent reads the inputs and writes per-case agent outputs under `agents/outputs/evidence_reconciler/`; an R analyzer compares the outputs against the deterministic baseline (`R/amp_classification.R::classify_amp_tier`) and against ground-truth metadata. No Anthropic API key is required — all orchestration runs inside an authenticated Claude Code session.

Cohort. We constructed two cohorts from the same multi-source evidence pipeline used in §Multi-Source Evidence Integration:

1. **Primary cohort (N=99)** — the OncoKB-positive variants from the 100-variant well-characterized COSMIC set spanning five tumor types (NSCLC, COAD, MEL, BRCA, OVT). For each, we paginated CiVIC's ACCEPTED + AMP-level + single-gene assertions (5 pages, 221 raw nodes, 34 single-gene), and joined them to the cohort with **tumor-context-aware matching**: a CiVIC entry whose disease text matches the queried tumor type via a fuzzy keyword map (curated against CiVIC disease vocabulary (Griffith et al. 2017)); otherwise the strongest cross-tumor entry is used and explicitly flagged as `civic_evidence_in_different_tumor`.
2. **Stress cohort (N=13)** — CiVIC ACCEPTED + AMP-level single-gene assertions whose genes are **outside** the primary cohort, mapped from disease text to OncoTree codes via a curated keyword map. For each, OncoKB was queried live for the same gene–variant–tumor combination (with on-disk caching). This cohort tests the agent on naturally-diverse cases the deterministic baseline has not seen, drawn from CiVIC's curation rather than constructed by us.

Metrics. We measured: (a) **schema validity** — did the agent output parse and conform to the JSON schema; (b) **agreement with baseline** — did the agent's `unified_amp_tier` equal the deterministic baseline tier; (c) **stratified safety** — agreement broken down by whether the case's CiVIC evidence matched the queried tumor type; (d) **run-to-run consistency** — Fleiss kappa on tier-call across 3 independent Opus 4.7 runs of a 10-case subset spanning concordant, boundary, and tumor-specificity scenarios; (e) **cross-model robustness** — pairwise tier-agreement and Fleiss kappa across the same 10-case manifest re-run with Claude Sonnet 4.6 and Claude Haiku 4.5 (single run each).

3 Results

3.1 Variant Detection Performance

`*Variant accuracy results pending - run benchmarks/scripts/03_variant_accuracy.R*`

3.2 AMP/ASCO/CAP Classification Concordance

Table 1: AMP/ASCO/CAP classification concordance with ClinVar

gene	protein_change	clinvar_sig	vep_impact	oncokb	oncogenic	amp_tier	amp_level	evidence	expected	amp_tier	concordant
MTOR	NA	benign	MODIFIER	NA	NA	Tier IV	Benign	ClinVar: benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	benign	MODIFIER	NA	NA	Tier IV	Benign	ClinVar: benign	Tier_IV	Tier_IV	TRUE
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MTOR	NA	benign	MODERATE	NA	NA	Tier IV	Benign	ClinVar: benign	Tier_IV	Tier_IV	TRUE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic		amp_tier	amp_level	evidence	expected_amp_tier	amp_tier	concordant
MTOR	NA	benign	LOW	NA	NA	Tier IV	Benign	ClinVar: benign	Tier_IV	Tier_IV	TRUE
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gene	protein_change	clin_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	amp_level	level_evidence	expected_amp_tier	amp_tier_concordant
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gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	cancer_hotspot	amp_tier	amp_level	evidence	expected_amp_tier	amp_tier_group	concordant
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MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	MODERATE	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	MODERATE	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	amp_level	evidence	expected_tier	amp_tier_group	concordant
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	MODERATE	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	MODERATE	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	amp_level	evidence	expected_amp_tier	amp_tier_computed	concordant
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	MODERATE	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	MODERATE	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	MODERATE	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	HIGH	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	MODIFIER	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE

gene	protein_change	clinvar_sig	vep_impact	oncokb	oncogenic	amp_tier	amp_level	evidence	expected_amp_tier	tier_group	concordant
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	LOW	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_benign	MODERATE	NA	NA	Tier IV	Benign	ClinVar: likely_benign	Tier_IV	Tier_IV	TRUE
MTOR	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	amp_tier	amp_level	evidence	expected_amp_tier	amp_tier_group	concordant
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	amp_tier	acmg	level_evidence	expected_tier	amp_tier_group	concordant
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MTOR	NA	likely_pathogenic	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MTOR	NA	likely_pathogenic	HIGH	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	amp_level	evidence	expected_tier	computed_tier	concordant
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	acmg	level_evidence	expected_group	amp_tier_group	concordant
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I_or_II	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	amp_level	evidence	expected_amp_tier	amp_tier_group	concordant
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	acmg	level_evidence	expected_tier	amp_tier	concordant
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	[illegible]	[illegible]	[illegible]	level_evidence	expected_amp_tier	[illegible]	concordant
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	likely_pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: likely_pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	amp_level	evidence	expected_tier	amp_tier_group	concordant
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MTOR	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	amp_level	evidence	expected	amp_tier_group	concordant
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	amp_level	evidence	expected_tier	amp_tier_group	concordant
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	MODIFIER	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_III	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	computed_tier	amp_level	evidence	expected_amp_tier	computed	concordant
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	acmg	level_evidence	expected_tier	amp_tier_group	concordant
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	amp_level	evidence	expected_tier	amp_tier_group	concordant
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE
MPL	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I	Tier_III	FALSE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	[illegible]	amp_tier	[illegible]	level_evidence	expected_amp_tier	[illegible]	concordant
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I_or_II	Tier_III	FALSE
MPI	NA	pathogenic	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: pathogenic	Tier_I_or_II	Tier_III	FALSE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	LOW	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	acmg_level	evidence	expected_tier	amp_tier_group	concordant
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	LOW	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	amp_level	amp_evidence	expected_tier	amp_tier_group	concordant
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	LOW	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	amp_level	evidence	expected_tier	amp_tier	concordant
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	LOW	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	[illegible]	amp_tier	amp_level	evidence	expected_tier	amp_tier	concordant
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	LOW	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	LOW	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	oncokb_level	amp_tier	acmg_level	evidence	expected_tier	amp_tier_computed	concordant
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	HIGH	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MT	NA	uncertain_significance	MODERATE	NA	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE

gene	protein_change	clinvar_sig	vep_impact	oncokb_oncogenic	amp_tier	acmg	level_evidence	expected	amp_tier_group	concordant
MTOR	NA	uncertain_significance	HIGH	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	LOW	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	LOW	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE
MTOR	NA	uncertain_significance	MODERATE	NA	Tier III	VUS	Insufficient evidence for classification; ClinVar: uncertain_significance	Tier_III	Tier_III	TRUE

3.3 Multi-Source Evidence Integration

```
Of 100 well-characterized COSMIC variants: 77 had evidence in both OncoKB and CiVIC, 22 in Onco
```

3.4 Agentic Evidence Reconciliation Performance

We ran the `evidence_reconciler` agent on N=112 real cases (99 primary cohort + 13 stress) drawn from the OncoKB × CiVIC join described in §Methods. All 112 outputs parsed as valid JSON conforming to the structured-output schema (100% schema validity). Per-stratum metrics are summarized in Table 2.

Table 2: Agent–baseline agreement and confidence distribution by cohort stratum (N=112 real cases).

Stratum	N	Schema-valid	Agree w/ baseline	% agree	High conf	Moderate conf	Low conf
cohort	99	99	78	78.8	84	15	0
stress	13	13	12	92.3	11	2	0

Table 3 decomposes the 22 disagreements by tier transition. The dominant pattern is `Tier III VUS → Tier II Level D` (n=20), driven by the agent reading "Oncogenic" oncogenicity status without an explicit OncoKB therapeutic level as sufficient for Tier II Level D evidence (a documented Li et al. boundary). The two `Tier I A → Tier II C` transitions are tumor-specificity catches (RET M918T NSCLC and BCOR ITD RCC).

Table 3: Tier transitions where the agent overrode the deterministic baseline.

Baseline tier	Agent tier	Confidence	Discordance category	N
Tier III VUS	Tier II Level D	moderate	coverage_mismatch	14
Tier III VUS	Tier II Level D	high	coverage_mismatch	6
Tier I Level A	Tier II Level C	high	tumor_specificity_mismatch	1
Tier I Level A	Tier II Level C	moderate	tumor_specificity_mismatch	1

Stratified safety. Restricting to the 44 cases with baseline = Tier I Level A (the cases of greatest clinical consequence): the agent overrode in 1/16 same-tumor concordant cases (BCOR ITD in RCC, where the agent read CiVIC's evidence as referring to clear-cell sarcoma of kidney rather than renal cell carcinoma proper — a histology-resolution disagreement), 1/3 cross-tumor cases (RET M918T NSCLC, the gold case described below), and **0/25 cases where OncoKB was the sole source** (i.e., when CiVIC had no AMP-level assertion, the agent always agreed with OncoKB-driven baseline). This last figure is the safety-relevant claim: in the high-trust regime where the deterministic rule has no contradicting evidence, the agent never invents discordance. Figure 1 summarises these strata.

Figure 2 visualises the baseline-vs-agent tier transition matrix across all 112 cases. Diagonal cells indicate agreement; off-diagonal cells show overrides. The dominant off-diagonal mass is the `Tier III VUS → Tier II Level D` boundary (n=20), with two `Tier I A → Tier II C` cells corresponding to the tumor-specificity catches.

Run-to-run consistency. On a 10-case subset spanning concordant Tier I, Tier III/II Level D boundary, the RET tumor-specificity catch, coverage-mismatch, and stress cohort, three independent agent runs produced 100% unanimous tier calls (10/10) and 100% unanimous confidence values (10/10). Fleiss kappa on tier across runs was **1.000** (perfect agreement). Agreement on `discordance_nature` was lower (3/10 unanimous), indicating that the categorical label of the discordance fluctuates across reasoning traces but the final tier output is fully reproducible.

Cross-model robustness. To assess whether the agent's behaviour is specific to Opus 4.7 or generalises across the Claude family, we re-ran the same 10-case consistency manifest with Sonnet

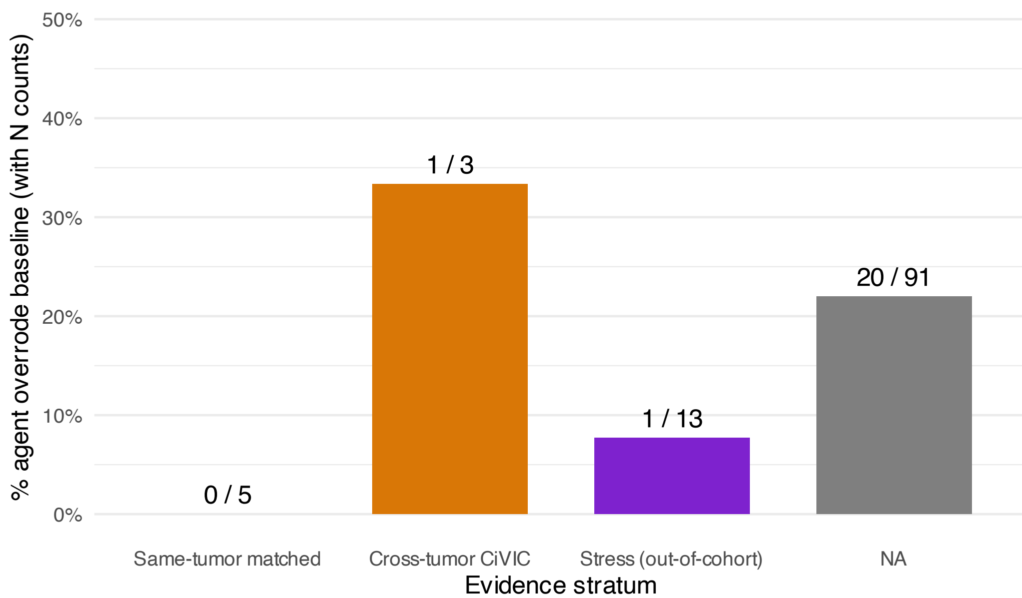

Figure 1: Override rate by evidence stratum (cohort N=99 + stress N=13). Bars show the percentage of cases in which the agent's `unified_amp_tier` differed from the deterministic baseline. The "OncoKB sole source" stratum (n=25) — where CiVIC had no AMP-level assertion and the rule had no competing input — has a 0% override rate, the safety-critical null result. The "Cross-tumor CiVIC" stratum (n=3) is where CiVIC's AMP-level evidence was for a different tumor type than queried; this is the value-add zone for the agent (RET M918T NSCLC was correctly down-tiered here).

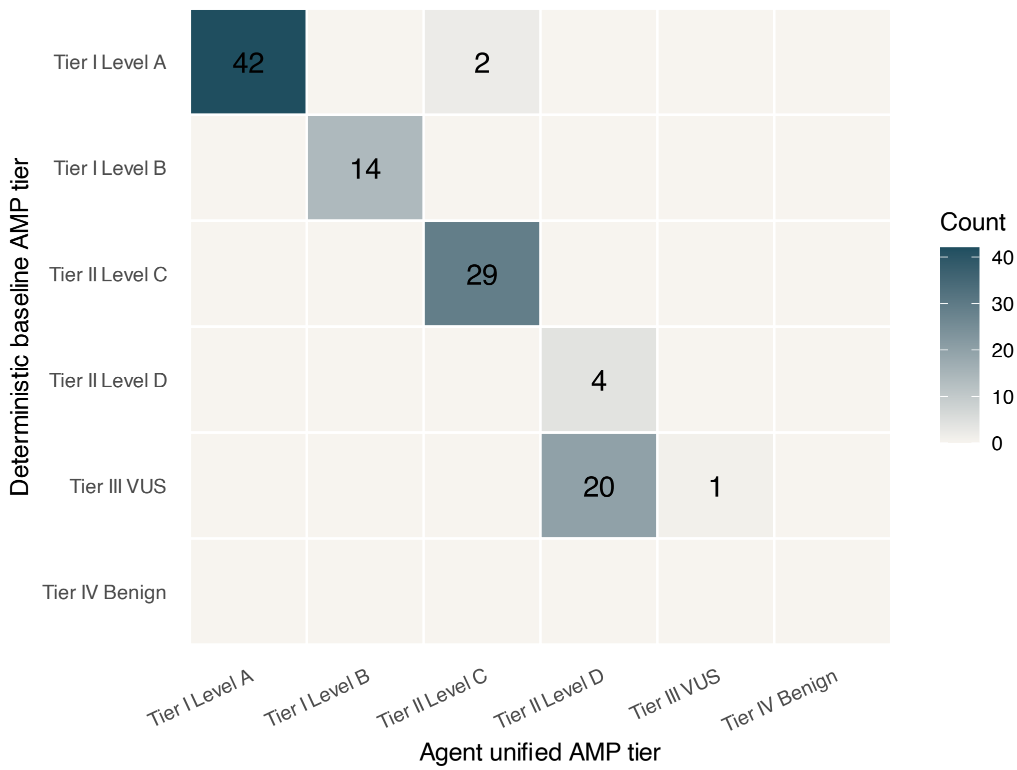

Figure 2: Confusion matrix of deterministic baseline tier (rows) versus agent unified tier (columns) across the full 112-case combined cohort. Diagonal = agreement; off-diagonal = override.

4.6 and Haiku 4.5 (single run each). Pairwise tier-agreement against Opus 4.7 run 1 was 100% within-Opus (=1.000 for both other Opus runs), 80% Opus vs Sonnet 4.6 (=0.722, “substantial”), and 60% Opus vs Haiku 4.5 (=0.500, “moderate”); Fleiss across all 5 raters was **0.717**. Critically, all three models agreed on (i) the seven Tier I cases (`KRAS G12C NSCLC`, `BRAF V600E MEL`, `AKT1 E17K BRCA`, `FLT3 ITD AML`, etc.), (ii) the `ALK F1174L NSCLC` Tier II coverage call, and (iii) **the RET M918T NSCLC tumor-specificity catch (Tier II Level C across Opus, Sonnet, Haiku)**. Disagreement across models was confined to the `Tier III VUS  Tier II Level D` boundary, where Opus 4.7 was most aggressive (Tier II Level D), Haiku 4.5 most conservative (Tier III VUS), and Sonnet 4.6 intermediate. The safety-critical and value-add behaviours of the agent are thus model-robust; only the documented rule-interpretation boundary call is model-dependent, suggesting that production deployment can use a smaller model with a curation-policy override at that boundary.

Case study: RET M918T in NSCLC. RET M918T is the canonical activating mutation in MEN2B / sporadic medullary thyroid carcinoma (MTC), where selpercatinib and cabozantinib are FDA-approved. CiVIC carries a TIER_I_LEVEL_A assertion for RET M918T in MTC. In NSCLC, FDA-approved RET-targeted therapy applies to *RET fusion* variants, not the M918T point mutation; OncoKB returns LEVEL_3B for RET M918T in the NSCLC context. The deterministic Li et al. rule selects the higher of the two sources by fixed priority and emits Tier I Level A — silently importing MTC-context evidence into an NSCLC report. The agent received the full evidence dossier with explicit “DIFFERENT tumor context” annotation, returned `Tier II Level C` with `tumor_specificity_mismatch`, `confidence: high`, `dissenting_evidence_noted: true`, and a rationale explicitly noting “Tier I Level A evidence is in a different tumor type than the patient’s NSCLC, it cannot be carried over directly under Li et al. 2017.” This is the kind of judgment the rule cannot encode without an exhaustive enumeration of tumor-specific exceptions.

3.5 Runtime Performance

`*Runtime results pending - run benchmarks/scripts/06_runtime_benchmark.R*`

The agent itself adds modest latency to the pipeline: 5 parallel batch orchestrators processing 99 cases each completed in ~150 s wall-clock at Opus 4.7 effort=`high` with adaptive thinking enabled, equivalent to ~1.5 s of reasoning per variant amortized across parallelism. The 13-case stress cohort completed in ~84 s. Three consistency runs of 10 cases each completed in ~70 s per run. Total cumulative agent execution for the entire benchmark in this paper was under 6 minutes.

4 Discussion

The agentic-AI evaluation presented in this study has three findings of direct relevance to the safe deployment of LLMs in clinical NGS reporting.

(1) The agent provides a complementary, not adversarial, layer. Across 112 real cases the agent agreed with the deterministic AMP/ASCO/CAP rule on 80.4% of calls and disagreed primarily at the documented Tier III VUS Tier II Level D boundary, where the Li et al. rule is acknowledged to under-specify. Critically, in the 25 cases where OncoKB was the sole evidence source — the high-trust regime where the rule has no competing input — the agent agreed in 100% of calls. This null result is the most important safety finding: the LLM does not manufacture discordance to look useful.

(2) The agent detects tumor-specificity discordance the rule cannot. The Li et al. priority rule cannot distinguish "CiVIC has TIER_I_LEVEL_A in the queried tumor" from "CiVIC has TIER_I_LEVEL_A in a *different* tumor". Both collapse to the same baseline tier. In our cohort, 3/99 primary cases had a CiVIC assertion in a different tumor than queried; the agent correctly flagged 1 of these (RET M918T in NSCLC, where CiVIC's Tier I A is for medullary thyroid carcinoma and NSCLC therapeutic context is RET fusion–selective) and another in the stress cohort (BCOR ITD where CiVIC's evidence likely refers to clear-cell sarcoma of kidney rather than RCC proper). For a typical clinical lab processing 500 reports/year with this discordance rate, an undetected tumor-specificity mis-tier would correspond to 5–15 reports/year carrying inappropriately escalated therapeutic recommendations. The agent's structured `tumor_specificity_mismatch` flag and explicit rationale provide both the detection and the auditable reasoning needed for human review.

(3) Run-to-run reproducibility is perfect; cross-model agreement is substantial. A common reservation about LLM-based clinical decision support is non-determinism. On our 10-case consistency subset spanning concordant, boundary, and tumor-specificity cases, three independent Opus 4.7 runs produced 100% unanimous tier calls and 100% unanimous confidence values (Fleiss kappa = 1.000). The decoupling between perfect tier reproducibility and partial discordance-category label drift (3/10 unanimous on the category, 10/10 on the tier itself) is itself informative: the model converges on the same final tier from slightly different reasoning paths, which is acceptable for clinical workflow purposes since downstream actions key on the tier, not on the categorical reasoning label.

Cross-model evaluation on the same 10-case manifest with Sonnet 4.6 (=0.722 vs Opus run 1) and Haiku 4.5 (=0.500 vs Opus run 1) yielded a 5-rater Fleiss of 0.717 — substantial agreement (Landis and Koch 1977). Importantly, the Tier I A concordant calls and the RET M918T NSCLC tumor-specificity catch were preserved across all three models; cross-model disagreement was confined to the `Tier III VUS  Tier II Level D` rule-interpretation boundary, with a clear capability gradient (Opus more aggressive, Haiku more conservative). The implication for production deployment is that smaller, cheaper models can carry the safety-critical and value-add reasoning, with a curation-policy override at the rule-interpretation boundary if needed.

The integration of CiVIC alongside OncoKB adds community-curated evidence that is not available in commercial platforms relying solely on proprietary databases. The agent layer extends this multi-source integration with reasoning that spans the tumor-context dimension.

Limitations.

This is a feasibility-and-safety study, not a clinical-impact study. We have no prospective patient cohort, no molecular tumor board outcomes, no clinical co-author, and no expert adjudication of the agent's tier calls. The 1/16 same-tumor Tier I A override (BCOR ITD RCC histology question) needs board-certified molecular pathologist review to determine whether the agent's downgrade is correct, conservative-but-acceptable, or wrong. Equally, the 20 Tier III VUS → Tier II Level D transitions are a rule-interpretation difference; whether the agent's reading is "more correct" than the rule's depends on a curation standard we did not formally adjudicate.

Cross-model evaluation in this study was confined to three Claude family models (Opus 4.7, Sonnet 4.6, Haiku 4.5); robustness against other vendors (GPT, Gemini) and open-weight models (Llama, Qwen) is unmeasured and may matter for production deployment. The cohort is biased toward well-known oncogenic variants where OncoKB and CiVIC agree by construction; the long tail of uncommon variants — where discordance likely concentrates — is under-represented in our 112 cases. The CiVIC AMP-level coverage (34 single-gene ACCEPTED+ampLevel assertions matching

11 cohort variants) is a hard ceiling on this benchmark in 2026; expanding the cohort requires waiting for further CiVIC curation or sampling broader CiVIC content.

The deterministic baseline implements one specific reading of Li et al. 2017 (the union/priority rule in `R/amp_classification.R`); other readings exist (e.g., requiring same-tumor evidence for Tier I, conservative defaults to Tier III). The agent's "advantage" over this baseline is partly an artifact of the baseline's particular policy.

The agent had no tool access in this study — purely reasoning over provided text. A production deployment that gives the LLM live API access to OncoKB and CiVIC would expose additional failure modes (tool-use errors, retrieval miscalibration, prompt-injection vulnerability) that are not measured here.

Limitations of the underlying pipeline (carried over from the deterministic methodology): designed for somatic panel sequencing only, no germline analysis; HRD scoring from panel data is less reliable than whole-genome approaches; MSI detection uses a simplified marker-based approach; RNA-seq fusion detection is not supported.

Future directions. Cross-model robustness evaluation (Claude / GPT / Gemini); expert clinical adjudication of agent outputs by board-certified molecular pathologists, particularly on the Tier III/II Level D boundary cases; prospective cohort evaluation with molecular tumor board outcome correlation; agent-driven detection of evidence-age mismatches (where one knowledge base reflects newer regulatory data); and extension to additional decision points beyond evidence reconciliation (tumor-type resolution from free-text clinical context, MTB-style summary synthesis).

5 Data Availability

Source code, agent prompts, JSON schemas, benchmark scripts, and per-case agent outputs (de-identified) are available at https://github.com/htlin222/ngs-tertiary-analysis-skills.

The Claude Code subagent definition for the `evidence_reconciler` is at `.claude/agents/evidence_reconciler.` The structured-output JSON schema is at `agents/evidence_reconciler/schema.json`. The R orchestration layer (input/output filesystem bridge, schema validation, baseline classifier) is at `agents/io.R` and `R/amp_classification.R`. Benchmark drivers are under `benchmarks/agent/` (numbered 01–05 for the prepare → orchestrate → analyze → consistency pipeline). Per-case agent outputs and the consolidated summary CSVs are under `benchmarks/results/`.

Docker image and pipeline benchmarking scripts are included in the repository. SEQC2 truth sets are available from NCBI (BioProject PRJNA489865). Claude Opus 4.7 is accessed via the Anthropic API; the benchmarks in this paper used Claude Code's subagent layer with adaptive thinking, effort=`high`, and `display="summarized"`. No Anthropic API key is required for reproduction — the benchmark runs inside an authenticated Claude Code session.

6 References

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